MIL-W-80187A
22 May 1985
SUPERSEDING
MIL-W-80187
8 September 1969

MILITARY SPECIFICATION

WELDING MACHINES, ULTRASONIC BONDING, MINIATURE WIRE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of welding machine for welding miniature wire to small electrical components by an ultrasonic process.

1.2 Classification. This specification covers ultrasonic welding machines of the following classes and sizes. The class and size to be furnished shall be as specified (see 6.2.1).

Class I - 20 kilohertz (kHz).
Class 11 - 40 kilohertz (kHz).
Class 111 - 60 kilohertz (kHz).
Class IV - 100 kilohertz (kHz).

Size 1 - 5watts.
Size 2 - 10 watts.
Size 3 - 50 watts.
Size 4 - 100 watts.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified. the following sr)ecifications. standards. and handbooks of the issue listed in that issu; of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3438

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SPECIFICATIONS

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NEMA ICS 1 - Industrial Control and Systems.

ANSI/NFPA 70 - National Electrical Code.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

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RESISTANCE tiELDER MANUFACTURERS ASSOCIATION (RWMA)

RWMA Bulletin Number 16 - Resistance Welding Equipment Standards.

(Application for copies should be addressed to the Resistance Welder Manufacturers Association, 1900 Arch Street, Philadelphia, pA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 De~ign. The welding machines shall be new, of the manufacturer's current design, and shall consist of all components required for the design functions. The design shall provide for the feeding of miniature wire through a capillary or welding head tip, the holding of the electrical component or unit to be worked, and the micropositioning of the head or the work. Controls for the ultrasonic welding process shall be provided. The design shall include provisions for the timed ultrasonic bonding of the wire to the work and the severance of the wire. A microscope shall be provided to observe the welding process. The design shall provide efficient and safe operation. All parts subject to wear shall be accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (S1) shall be used in the design and construction of the welding machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to S1 units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units S) or the International System of Units (S1) shall be-used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the welding machine shall be graduated in the same system.

3.2.1.1.1 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the S1 System of

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Measurements. When a dual US and S1 system is furnished, both dials shall have independent zero adjustments and both dials shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The welding machine may contain reclaimed materials provided such materials will not jeopardize the welding machines intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the welding machine.

3.2.3 Energy efficiency. The welding machine and its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the Industry.

3.2.4 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the welding machine that present safety hazards. The safety devices shall not interfere with the operation of the welding machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All welding machine parts, components, mechanisms, and assemblies furnished on the welding machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the welding machine itself. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

3.2.6 Mercury restriction. The welding machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the welding machine.

3.2.8 Environmental protection. The welding machine shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the welding machine shall not emit materials hazardous to the ecological system as prescribed by federal, state, or local statutes in effect at point of installation (see 6.4).

3.2.9 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or welding machine.

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3.3 Const~uction~' The welding machine shall be constructed of parts which are new. without defects, and free of repairs. The structure shall be capable of wi~hstanding all forces encountered-during operation of the welding machine at its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the welding machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed In a manner to Prevent change of tightness. Fastening devices subject to removal or adjustment shall ~ot be swaged, peened, stak~ed, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the welding machine shall be painted in accordance with the manufacturer's commercial practice, provided that all surfaces to be painted shall be cleaned of all foreign matter and at least one coat of primer and one coat of finish color is applied.

3.3.6 Threads. All threaded parts used on the welding machine and its related attachments and accessories shall conform to FED-STD-H28.

3.4 Components. The welding machine shall consist of not less than the ultrasonic frequency converter, the ultrasonic welding head, the workholding unit, the wire handling unit, the micropositioning unit, the optical unit, the controls, and the enclosures.

3.4.1 Ultrasonic frequency converter. The machine shall be furnished with an ultrasonic frequency converter. The ultrasonic frequency converter shall consist of transformers, rectifiers, and other components as required to provide the necessary welding voltage, current, and frequency characteristics. All components shall be securely fixed and mounted to eliminate

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lamination shifting from shock, vibration, or other conditions encountered during service, operation, shipping, and storage. Transformer insulation shall be in accordance with RWMA Bulletin Number 16 for class A or better insulation.

3.4.1.1 Ultrasonic power output. The frequency converter shall provide ultrasonic weldlng power output as required for the class and size machine specified. Adjustment of the power output shall be provided by digital display controls. The power output of the converter shall not vary more than 1.0 percent from setting. A power indicating meter shall be mounted in the front panel. Unless otherwise specified, a single channel power output shall be furnished. When specified a dual channel power output shall be furnished (see 6.2.1).

3.4.1.2 Transducer-coupling assembl~. The transducer-coupling assembly shall be powered at a fixed frequency by the frequency converter. An indicating meter shall be incorporated to facilitate adjustment of nominal operating frequency.

3.4.1.3 Weld pulse duration timer. Each power channel shall have a weld pulse timer giving a weld pulse duration range from a minimum of not more than 0.05 seconds to a maximum of not less than 0.20 seconds. Pulse duration repeatability shall be within ~0.01 second. All timers shall be standardized at 0.075 seconds +0.002 seconds. Pulse duration shall be adjustable by the digital displ;y control. A means of continuously firing each channel shall be provided for set up purposes.

3.4.1.4 Biasing current. Transducer biasing current, when required, shall be delivered to the transducer through the high frequency power connections of the frequency converter. Biasing or polarizing current to the transducer shall be regulated as required for the performance specified.

3.4.1.5 Test points. The frequency converter shall incorporate test points for the measurement and calibration of output voltage, current, frequency, and pulse duration.

3.4.2 Ultrasonic welding head. The ultrasonic welding head shall include a welding tip and a transducer coupling assembly consisting of a transducer, a coupler, and a mounting. When the welding head is not attached to the main enclosure or cabinet, flexible connecting cables and hoses shall be provided. The head shall have facilities for the application of an adjustable and repeatable ultrasonic welding force. The time and force shall be repeatable to within ~10.0 percent of the preselected adjustment. The force shall be independent of the operation of the micropositioner. When a two-stage force system is provided, these requirements shall apply to both stages.

3.4.2.1 Welding tip. The welding tip shall be mounted on the coupler. The welding tip shall position the wire and transmit the vibratory welding

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energy and force to the weldment. Unless otherwise specified (see 6.2.1), wire size and other characteristics of the welding tip shall be the manufacturer's standard design.

3.4.2.2 Transducer coupling assembly. The transducer coupling assembly shall have a resonant frequency for the class specified. The matching of frequency characteristics shall be within ~10.0 percent. Frequency tolerance of the system shall not exceed :1.0 percent of nominal values specified.

3.4.2.3 Transducer. The transducer shall convert high frequency electrical energy to mechanical vibratory energy of the same frequency. The transducer shall be rigidly attached to the coupler in a manner to minimize acoustic wave distortion or mechanical vibration during transmission to the coupler. The transducer material shall prevent performance deterioration from aging or overheating.

3.4.2.4 Coupler. The coupler, including the mechanical transformer if applicable, shall transmit ultrasonic energy to the welding tip. The coupler shall be constructed geometrically to provide for the transmission of the energy as required for the performance specified herein. The construction of the coupler shall prevent the generation of undesirable acoustical characteristics.

3.4.2.5 Coupler mounting. The coupler shall be mounted in such a manner as to isolate the ultrasonic system from the machine frame. The mounting shall be rigid to prevent detrimental mechanical deflection of the welding tip when the normal range of static clamping force is applied.

3.4.3 Work holding unit. The work holding unit shall be constructed of materials that are compatible with ultrasonic welding processes, and shall consist of an anvil, base, header, or other suitable device for holding the work. The work holding unit shall prevent transmission of ultrasonic ener- The unit shall hold a flat, square workpiece with side dimensions up to ~~nches, and thickness dimensions up to 0.5 inch without damage to the workpiece or weldment. The clamping force range shall be sufficient to hold the workpiece stationary during the welding process.

3.4.4 Wire handling unit. The wire handling unit shall consist of the wire reel unit, the w~re feeding unit, and the wire cutoff unit.

3.4.4.1 Wire reel unit. The wire reel unit shall support the wire reel, spool, or co1l for wire storage and feeding. The wire reel size shall be compatible to the welding tip and wire size specified in 3.4.2.1.

3.4.4.2 Wire feeding unit. The wire feeding unit shall feed the wire without deformation from the reel unit, to the welding tip. When furnished, demand type feed rate shall be as standard with the manufacturer. Unless otherwise specified, the wire feed motion shall be controlled manually either by a hand or foot actuator. When specified the wire feed shall be motorized, with feed rates and wire size range as specified (see 6.2.1).

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3.4.4.3 Wire cutoff unit. The wire cutoff unit shall utilize scissors, flame, or a wedge-puller unit. Unless otherwise specified (see 6.2.1), a scissors type cutoff unit shall be furnished.

3.4.5 Micropositioning unit. The micropositioning unit shall provide relative micropositioning between the welding head and the work so that ultrasonic welding may be performed within the boundary of a cube with a base 1.0 inch square and .050 inch high. Micropositioning shall be accomplished by movement of the welding head in the "Y" plane and the workpiece in the "Z" plane, or by any other combination of movements. Precise movement shall be within .000025 inch and shall be observed by an optical microscope unit. Micropositioning shall be accomplished by the operator in his normal operating position. When a micropositioner is required on the wire cutting unit or on the wire feeding unit, the precision and space of motion shall be compatible with performance of functions as specified herein.

3.4.6 Optical unit. The optical unit shall be movable away from the work area and shall be capable of remaining in focus with the work. The optical unit shall consist of a stereo binocular microscope type instrument, through which the operator can view the welding operation when in his normal operating position. Magnification shall be variable from a lowest value of at least 15 power to a highest value of at least 60 power at a nominal working distance of 4 inches. An illuminator of the standard type normally furnished by the manufacturer for the class and size of machine specified shall be furnished.

3.4.7 Controls. The controls shall provide complete control of variables in all necessary functions of the machine. A weld pulse initiation and channel selector shall be provided. The weld pulse shall not be initiated until full bonding pressure is established and any force induced vibrations have decayed in the transducer coupling assembly. The frequency converter shall be provided with a motor and a 115 volt AC nominally closed switch capable of handling one ampere to lock out machine cycling during weld duration. Each control and indicator shall be clearly and legibly marked for function and identification. The machine shall be completely wired through the controls and ready for connection to the applicable power source. Unless otherwise specified (see 6.2.1), all controls and auxiliary circuits shall be in accordance with ANSI/NEMA ICS 1.

3.4.8 Enclosure. The enclosure shall be constructed of either corrosion resistant material or metal treated to resist corrosion. The enclosure shall have sufficient strength and be arranged to support all components. Ventilation features shall be adequate to prevent excess temperature under normal operating conditions of rated power output.

3.4.9 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

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3.4.10 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA 70. Each welding machine shall be wired to ~raw all of its electrical power from a 115 volt; single phase, 60 Hz circuit. Each welding machine shall have a fused safety disconnect switch or circuit breaker. An identified terminal for grounding the welding machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA 70 for the disconnect fuse rating.

3.4.10.1 Motors. Motors shall be rated for continuous duty and shall have ball or r-bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a dripproof enclosure.

3.4.10.2 Solid state components. Solid state design shall be used throughout for electronic components.

3.4.10.3 Control circuit voltage. Auxiliary control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage no greater than 24 volts.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the welding machine shall be not less than those specified for the welding machine size specified.

3.6 Performance. The welding machine shall be capable of ultrasonically welding miniature wire connections to electrical components. The welding machine shall weld the formed end of the wire to the electrical component by a combination of applied clamping force and timed ultrasonic vibratory motion. Each machine shall perform at not less than the frequency and power listed for the class and size covered in 1.2. The quality of the weldment shall be such, that under 30 power magnification no cracks are visible, and the combined lengths of voids shall not be greater than 10 percent of the interface length. Material deformation shall not be greater than 50 percent of the thickness of the affected material, or 35 percent of the total thickness of both materials. Wire weld deformations shall not be less than 5 percent and not greater than 60 percent reduction of total wire diameter. The breaking strength of the weld when subjected to shear, shall not be less than 35 percent of the weakest parent metal strength. Each machine shall meet all requirements of this specification when operating in an ambient air temperature between 70 degrees F and 85 degrees F. Welding components shall not be rated at less than 50 percent duty cycle. The welding machine shall perform the secondary functions of work-holding, wire-feeding, wire cutting, micropositioning of the head or the work, and stereo-microscopic viewing of the welding operation.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

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3.7 Standard equipment. Unless otherwise specified (see 6.2.1), all standard equipment normally furnished by the manufacturer, shall be furnished.

3.8 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.9 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

3.10 Marking on plates, charts, and decals. All words on plates, charts, and decals shall be in the tnglish language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. When these methods are not practical, the instructions shall be applied directly onto the item by environmentally protected decalcomania transfer, metal wrap-around tag, stencil, silk screen, or by any other method suitable for the use intended which shall be as permanent as the normal life expectancy of the item to which it is applied and capable of withstanding the environmental test and cleaning procedures specified for the item.

3.10.1 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each welding machine. The nameplate shall contain the information listed below. If the welding machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Ultrasonic power characteristics
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.12 Workmanship Workmanship of the welding machine and accessories shall be of a qua lty equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the perform-

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ante of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests in 4.6. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item Prior to beina offered for acce~tance under the contract. Unless otherwise specified (s~e 6.2.1), quality' conformance inspection shall consist of the examination in 4.5, the test in 4.6.1, and the inspection in 4.7. Failure of the item to pass the examination, the test, and the inspection shall be cause for rejection.

4.5 Examination. The welding machine shall be visually examined to determine compliance with all requirements of this specification.

4.6 Tests.

4.6.1 Operational test. The welding machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.6.2 Performance test. The welding machine shall be prepared for welding in accordance with the manufacturer's instructions. The welding machine shall perform ten welding cycles, producing actual welds to test for conformance with the performance requirements of 3.6. All ten specimen welds shall be examined at both magnification of 10 power and magnification of 30 power. Under the 10 power magnification, the welds shall show no evidence of cracks or voids. Under the 30 power magnification, the welds shall conform to the requirements of 3.6. Five specimen welds shall be tested for shear strength, and the breaking strength shall conform to the requirements of 3.6.

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4.6.3 Ultrasonic characteristics test. Ultrasonic probe measurements on the work holder shall show vibratory amplitude to be down at least 10 to 1 from welding-tip amplitude. Resonant frequency shall be established with the transducer coupling assembly unloaded and with a dummy welding tip installed.

4.6.4 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.4.9.

4.7 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Packaging, packing, and marking. Unless otherwise specified, packaging, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-W-45562.

6. NOTES

6.1 Intended use. The equipment covered by this specification is intended for precision ultrasonic welding applications.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the follow~ng:

a. Title, number, and date of this specification.

b. Class and size required (see 1.2).

c. First article inspection, if required (see 3.1).

d. If welding machine is required to be configured in a specific measurement system (US or S1), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or S1), state required system (see 3.2.1.1).

f. Dual calibrations (US and S1), if required (see 3.2.1.1.1).

g. Additional safety and health requirements, if required (see 3.2.5).

h. Painting, if different (see 3.3.5).

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i. Du~l channel power output, if required (see 3.4.1.1).

j. Wire size and other characteristics of the welding tip, if, different (see 3.4.2.1).

k. Motorized wire feed, if required (see 3.4.4.2).

1. Wire cutoff unit, if different (see 3.4.4.3).

m. Controls and auxiliary circuits, if different (see 3.4.7).

n. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.4.9).

o. Electrical system and input power, if different (see 3.4.10).

P* Motor enclosure, if different (see 3.4.10.1).

q= Size and capacity, if different (see 3.5).

r. Additional performance features, if required (see 3.6.1).

s. Standard equipment, if different (see 3.7).

t. Specify and fully describe optional equipment, if required (see 3.8).

u. Specify applicable repair parts, if required (see 3.9).

v. Nameplate, if different (see 3.10.1).

w. First article inspection, if different (see 4.3).

x. Quality conformance inspection, if different (see 4.4).

Ye If packaging, packing, and marking in accordance with MIL-W-45562 is-required, specify level required (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operators manuals, parts llsts, wlrlng diagrams, foundation and anchor bolt plans and acceptance test reports should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

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6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the welding machine itself. It does not require compliance with those OSHA requirements that concern "the welding machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual machine in the environment) the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional welding machine requirements that will integrate the new welding machine into its future operating environment. The welding machine shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial welding machines supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required, as in most cases, the guard configuration is dependent on the size and configuration of the workplaces. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1 (9)*

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.6 Classification cross reference. The classes and sizes of welding machines covered under the previous edition of the specification have been maintained in this revision with no change.

Custodian:
Air Force - 99

Review Activities:
Air Force - 84
DLA- GS

User ActivitY:
Navy - MC

Preparing Activity:
DLA - 1P

(Project 3438-0051)

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